

The Politics of Technological Change

Instructor: Alexander Trubowitz

Email address:

Office Hours: TBA

Overview: This course introduces undergraduate students to the study of technological change as a political process. How does politics affect the speed and direction of technological change? Is there one best technological path or does the optimal technology depend on the institutional environment? How do geopolitical rivalry, worker power, financial institutions, and competition between firms affect innovation? When do governments encourage technological catch-up and when do their interventions protect inefficient firms and impede private entrepreneurship? How can developing countries make the difficult transition from imitation to innovation?

The course is structured so that students are gradually exposed to theoretical tools for understanding the political economy of innovation while also learning the history of technological change since the Industrial Revolution. It proceeds from the early nineteenth century to the present, beginning with competing explanations for Britain's industrialization and ending with discussions of China's rise as a technological power, the politics of green innovation, and the sources of American leadership in radical innovation. Readings are drawn from economic history, political economy, labor sociology, international relations, and development economics and encompass classic contributions from Marx and Schumpeter alongside contemporary empirical research. By the end of the course, students will be equipped to conduct research on the politics of technological change and to evaluate contemporary debates about innovation using theoretically informed arguments and historical evidence.

Requirements:

Weekly reading: Students are expected to complete the week's readings before the first lecture of the week. The assigned material amounts to 70–90 pages per week, involving excerpts from longer scholarly works.

In-class midterm (Week 7): multiple choice, short answer, and essay. 30% of course grade.

In-class final exam: multiple choice, short answer, and essay. 30% of course grade.

Short paper (Week 8): Did government intervention promote or impede innovation in American manufacturing in the nineteenth century? Which interventions mattered most? This will require independent research focused on recent empirical studies of American industrial development in the nineteenth century. Be sure to critically assess the alternative position. 5–7 pages, double spaced. 25% of your grade, based on the written paper and a meeting with me following its submission. More detailed guidelines circulated separately.

Participation: active participation in class based on substantive engagement with the readings is expected and is worth 15% of your course grade. In this course, we will uphold the

university's code of conduct and values regarding open intellectual inquiry and equal respect for all class members.

Prerequisites: One prior course in the political science or economics department.

Electronic devices: Laptops are permitted for note-taking and for referring to course readings. Please be conscious of the fact that browsing the internet during class is distracting for the students sitting around you. If you plan to browse the internet during class time, expect to be called upon!

Plagiarism: Plagiarism is a serious breach of the University's code of conduct. It is assumed that you have carefully reviewed the University's policy on plagiarism. Your use of AI must adhere to the university's policy. Please note that you are unlikely to succeed on the midterm and final exam if you have not studied the readings in detail and thought about them yourself. Likewise, you need to be able to display ownership over the ideas expressed in your paper in our one-on-one meeting to receive a high grade. This will be considerably easier if you have written the paper yourself.

Accessibility: The University is committed to equal access for students with disabilities. Students with disabilities who have been approved to receive academic accommodations by the University must contact me as soon as possible so that the relevant accommodations can be made.

Week 1: Introduction. Why is technological change inherently political? How can new technologies disrupt political power and when do governments have an incentive to interfere with innovation?

Acemoglu, Daron, and James Robinson. "Economic backwardness in political perspective." *American Political Science Review* 100.1 (2006): 115-131.

Week 2: Labor costs, labor constraints, and new technologies (2 lectures). This week focuses on Britain's industrialization in comparison to France's trajectory in the early nineteenth century. How do labor costs and worker power affect incentives and opportunities to introduce labor-saving technologies? When might labor-intensive methods be preferable?

Allen, Robert C. "Why the industrial revolution was British: commerce, induced invention, and the scientific revolution." *Economic History Review* 64.2 (2011): 357-384.

Horn, Jeff. *The Path Not Taken: French Industrialization in the Age of Revolution, 1750-1830*. Cambridge, MA: MIT Press, 2006 (selections).

Sabel, Charles, and Jonathan Zeitlin. "Historical alternatives to mass production: politics, markets and technology in nineteenth-century industrialization." *Past & Present* 108 (1985): 133-176 (selections).

Week 3: International threats and incentives to innovate (2 lectures). This week introduces a different set of problems to the study of British industrialization. How does

geopolitical rivalry affect governments' incentives to promote innovation and overcome barriers to change? When does geopolitical competition instead encourage short-term behavior and the diversion of resources from productive uses?

Mokyr, Joel. "Cardwell's Law and the political economy of technological progress." *Research Policy* 23.5 (1994): 561-574.

McNeill, William. *The Pursuit of Power: Technology, Armed Force and Society Since A.D. 1000*. Chicago: University of Chicago Press, 1982 (selections).

Juhász, Réka. "Temporary protection and technology adoption: Evidence from the Napoleonic blockade." *American Economic Review* 108.11 (2018): 3339-3376.

Week 4: Technological change, de-skilling, and re-skilling (2 lectures). This week focuses on the labor market effects of technological change in Britain over the nineteenth century. How does technological change affect workforce skill requirements? How might worker power affect whether technological change deskills the workforce?

Marx, Karl. *Capital: Critique of Political Economy, Vol. 1*. Translated by Paul Reiter. Princeton: Princeton University Press, 2026. Chapter 14, "Division of Labor and Manufacture," and Chapter 15, "Machinery and Modern Industry."

Zeitlin, Jonathan. "Craft control and the division of labour: engineers and compositors in Britain 1890–1930." *Cambridge Journal of Economics* 3.3 (1979): 263-274.

More, Charles. *Skill and the English working class, 1870-1914*. London: Croon Helm, 1980 (selections).

Week 5: Incumbents, entrants, and innovation (2 lectures). This week focuses on the changing nature of the innovation process during the 'Second Industrial Revolution' of the late nineteenth and early twentieth centuries. Why might competition actually get in the way of research and development and when are established incumbent firms in a better position to innovate? How does the structure of the financial system affect the amount and distribution of innovative activity between incumbents and new entrants?

Schumpeter, Joseph. *Capitalism, Socialism, and Democracy*. London: Routledge, 1942 (selections).

Mowery, David, and Nathan Rosenberg. *Paths of Innovation: Technological Change in 20th-Century America*. Cambridge: Cambridge University Press, 1999 (selections).

Fohlin, Caroline. *Mobilizing Money: How the World's Richest Nations Financed Industrial Growth*. Cambridge: Cambridge University Press, 2011 (selections).

Week 6: Industrial policy, intellectual property, and innovation (2 lectures). This week focuses on the role of industrial policy and intellectual property rights in technological diffusion over the nineteenth century. When can states support technological diffusion

through industrial policy and when do their efforts get in the way? Is robust intellectual property protection necessary to encourage innovation and how do these protections affect patterns of technological diffusion?

Chang, Ha-Joon. *Kicking Away the Ladder: Development Strategy in Historical Perspective*. London: Anthem, 2002 (selections).

Juhász, Réka, and Claudia Steinwender. “Industrial policy and the great divergence.” *Annual Review of Economics* 16.1 (2024): 27-54.

Moser, Petra. “How do patent laws influence innovation? Evidence from nineteenth-century world’s fairs.” *American Economic Review* 95.4 (2005): 1214-1236.

Week 7: Limits to technological diffusion (2 lectures). This week focuses on the development and diffusion of American mass production methods in the early-to-mid twentieth century. How do economic and military considerations give governments an incentive to foster diffusion? How do institutions affect whether firms can actually integrate technologies from abroad, and why might firms choose to integrate foreign technologies only selectively?

Landes, David. *The Unbound Prometheus: Technological change and industrial development in western Europe from 1750 to the present*. Cambridge: Cambridge University Press, 1969 (selections).

Tolliday, Steven. “The Diffusion and Transformation of Fordism: Britain and Japan Compared.” *Between Imitation and Innovation: The Transfer and Hybridization of Productive Models in the International Automobile Industry*, ed. Robert Boyer et al., 57-95. Oxford: Oxford University Press, 1998.

Link, Stefan. *Forging Global Fordism: Nazi Germany, Soviet Russia, and the Contest over the Industrial Order*. Princeton: Princeton University Press, 2020 (selections).

Week 8: Military-industrial links and innovation (2 lectures). This week focuses on the role of the US military and the national security state in promoting the development of new technologies over the second half of the twentieth century. Did the demands of the military play a constructive role or did they distort the allocation of research and development away from its optimal uses?

Weiss, Linda. *America Inc.? Innovation and Enterprise in the National Security State*. Ithaca: Cornell University Press, 2014 (selections).

Melman, Seymour. “From Private to State Capitalism: How the Permanent War Economy Transformed the Institutions of American Capitalism.” *Journal of Economic Issues* 31.2 (1997): 311-330.

Heinrich, Thomas. “Cold war armory: military contracting in Silicon Valley.” *Enterprise & Society* 3.2 (2002): 247-284.

Week 9: Catch-up growth and the technological frontier (2 lectures). This week focuses on the challenges of promoting innovation in the late twentieth and early twenty-first centuries. How might the institutions most favorable to catch-up growth differ from those most conducive to innovation on the technological frontier? Why have countries found it difficult to imitate American institutions that are thought to promote radical innovation?

Acemoglu, Daron, Philippe Aghion, and Fabrizio Zilibotti. “Distance to frontier, selection, and economic growth.” *Journal of the European Economic association* 4.1 (2006): 37-74.

Anchordoguy, Marie. “Japan’s software industry: a failure of institutions?” *Research Policy* 29.3 (2000): 391-408.

Casper, Steven. *Creating Silicon Valley in Europe: Public Policy Towards New Technology Industries in Comparative Perspective*. Oxford: Oxford University Press, 2007 (selections).

Week 10: Automation and deindustrialization (2 lectures). This week focuses on the technological causes of structural economic change in the late twentieth and early twenty-first centuries. To what extent is the decline in industrial employment in the United States explained by technological innovations in manufacturing? Is the decline of manufacturing simply a consequence of technological advance or might deindustrialization itself erode American capacities for frontier innovation?

Fort, Teresa, Justin Pierce, and Peter Schott. “New perspectives on the decline of US manufacturing employment.” *Journal of Economic Perspectives* 32.2 (2018): 47-72.

Autor, David, and David Dorn. “The growth of low-skill service jobs and the polarization of the US labor market.” *American Economic Review* 103.5 (2013): 1553-1597.

Berger, Suzanne. *Making in America: From Innovation to Market*. Cambridge, MA: MIT Press, 2013.

Week 11: From imitation to innovation (2 lectures). What explains China’s rise from a site of cheap manufacturing to the home of frontier innovation? Does China’s success reflect central state direction or decentralization to local and regional governments? How does the Chinese method of promoting innovation differ from the American one?

Ang, Yuen Yuen. *How China Escaped the Poverty Trap*. Ithaca: Cornell University Press, 2016 (selections).

Wei, Shang-Jin, Zhuan Xie, and Xiaobo Zhang. “From “made in China” to “innovated in China”: Necessity, prospect, and challenges.” *Journal of Economic Perspectives* 31.1 (2017): 49-70.

Wang, Dan. *Breakneck: China’s Quest to Engineer the Future*. New York: W.W. Norton, 2025 (selections).

Week 12: Innovation and the green transition (2 lectures). Why have different countries excelled in different types of green technologies? When can states effectively promote the development of these technologies and when does state intervention backfire?

Nahm, Jonas. *Collaborative Advantage: Forging Green Industries in the New Global Economy*. New York: Oxford University Press, 2021 (selections).

Thurbon, Elizabeth et al. *Developmental Environmentalism: State Ambition and Creative Destruction in East Asia's Green Energy Transition*. New York: Oxford University Press, 2023 (selections).

Juhász, Réka, and Nathan Lane. "The political economy of industrial policy." *Journal of Economic Perspectives* 38.4 (2024): 27-54.

Week 13: Silicon Valley and American technological leadership (2 lectures). This week studies the causes of American leadership in computing and artificial intelligence. What explains Silicon Valley's remarkable success and how has the political economy of radical innovation in the United States changed over time? How does intensifying geopolitical competition affect technological development in the United States?

Saxenian, AnnaLee. *Regional Advantage: Culture and Competition in Silicon Valley and Route 128*. Cambridge, MA: Harvard University Press, 1994 (selections).

Soskice, David. "The United States as Radical Innovation Driver: The Politics of Declining Dominance?" In *The American Political Economy: Politics, Markets, and Power*, ed. J. Hacker et al., 323-350. Cambridge: Cambridge University Press, 2021.

Miller, Chris. *Chip War: The Fight for the World's Most Critical Technology*. New York: Scribner, 2022 (selections).